QGIS Plugin

LRS-Editor

For Linear Reference Systems (LRS)

Manual

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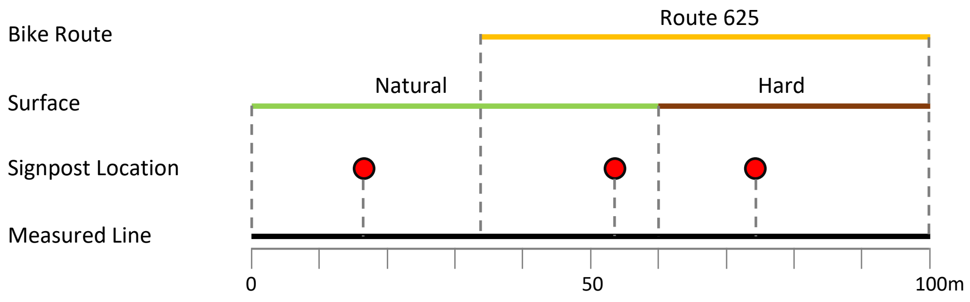
28.05.2026	1.4.0	Translation of the manual
07.07.2026	1.5.0	Migration QGIS 4

1. Introduction

1.1. What is Linear Referencing?

Linear referencing is the method to locate the relative position of objects along a measured line. The position of the objects is specified with distances from the starting point of the line:



The position of the object is stored in an attribute table without geometry. Therefore only one basic line geometry is required, which does not need to be subdivided as soon as the attribute value of an object changes.

A **Linear Reference System (LRS)** comprises all elements for managing the desired information as so-called **Events**[1)] along the basic line geometry, which is used to reference the information.

An LRS does not relate to a specific topic, but can be used for the management of various linear information (roads, public transport, non-motorized traffic, water system, etc.).

1.2. QGIS Plugin

The LRS-Editor as a QGIS plugin provides functions for setting up a Linear Reference System and editing tools for managing the Events in QGIS. The linear information is edited by points - so-called Event Points - directly in the QGIS map along the line, without having to manually manage measured values in a table. The LRS-Editor calculates the distances and enables efficient and correct tracking of events.

The dynamic visualization of events is also integrated in the LRS-Editor by adding and displaying the metric attribute values as segmented line layers (Views) in QGIS.

After installing the plugin, the following components are created:

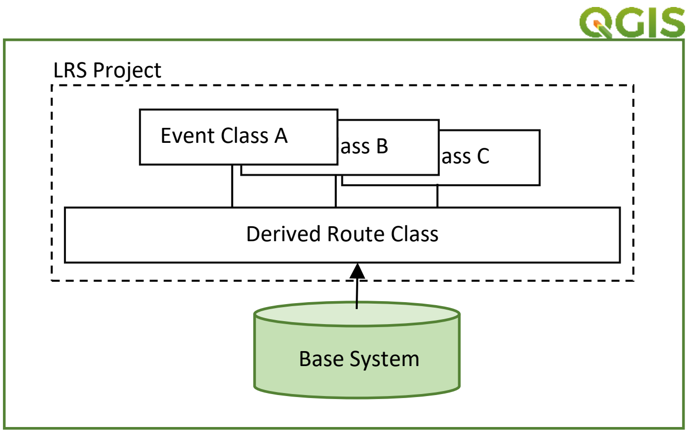
- Menu *LRS-Editor* (Plugins)
- *LRS-Editor Toolbar* with editing tools
- Docking widgets for editing the Events (not visible after installing)

[1)] The term ‘Event’ originates from ESRI's concepts for implementing linear referencing in its products.

2. Linear Referencing System

The linear reference system of the LRS-Editor comprises the following elements:

- **Base System** with the underlying line geometry
- **Route Class** derived from the Base System to generate the measured values of an Event
- Custom **Event Classes** with information as Events that reference the routes of the Route Class



An LRS is set up as a project in QGIS and corresponds to the desired topic. A QGIS project enables the administration of an LRS project.

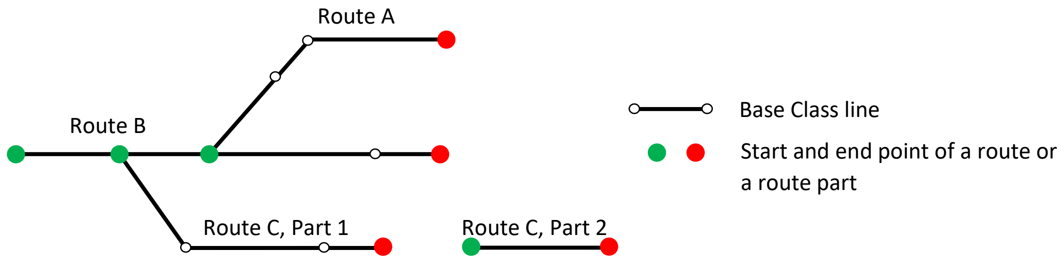
Supported is the PostgreSQL database system with PostGIS extension.

The Event Classes of an LRS project are managed together with the system tables in the same database schema by the LRS-Editor. The tables of the Base System can be kept in a different database, managed by a separate QGIS connection.

3. Base System

The Base System forms the basis of a linear reference system and contains an associated line geometry and points for generating the Route Class. The information of the following classes is managed by the Base System:

- **Base Class** with the route ID
- **Point Class** with the route ID, sort number of the route part and point type



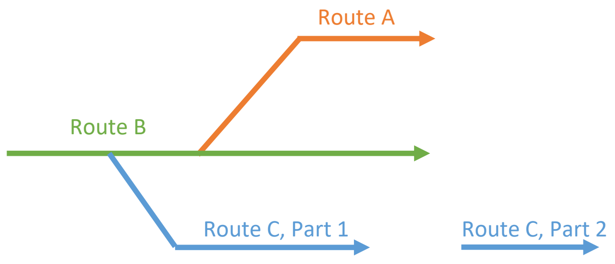
The Base Class contains the lines from which the LRS-Editor automatically creates the routes of the Route Class. The Point Class is used to define the start and the end and therefore the direction of a route or part of a route. The additional sorting number in the point class enables the creation of routes that consist of several independent parts.

When creating the Route Class, the LRS-Editor checks the topology of the Base System and outputs information in a log file. Routes that cannot be created correctly due to an incorrect topology are not transferred to the Route Class. In the same process changes to a line in the Base Class are automatically passed on to the route. The Route Class is finally updated.

Base and Point Class are managed outside the LRS-Editor by the data owner using the usual GIS tools in QGIS. They can be stored in a different database from the LRS project in PostgreSQL.

4. Route Class

The Route Class contains the – from the Base System – generated, directed routes and forms the basis for referencing Events of an Event Class. The LRS-Editor uses the routes to derive the measured values of the Event Points and manages this information in the database. After manually made changes of the Base Class in QGIS, the LRS-Editor refreshes the routes in a update process, and the changes are automatically passed on to the Event Classes with subsequently verification by the user.



Directed routes generated from the Base Class and their parts

The Route Class is added to the QGIS project as a layer; it is read-only and cannot be edited.

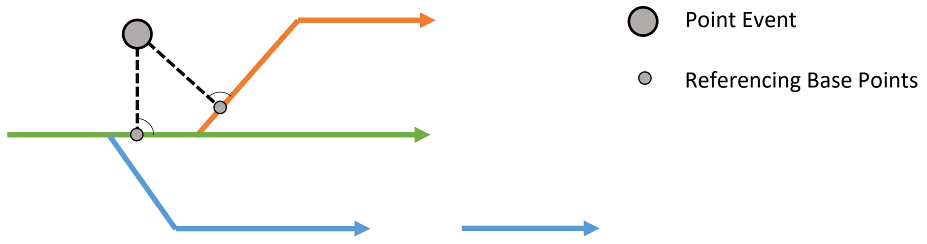
5. Event Classes

With the LRS-Editor following types of Event Classes can be created and managed:

- **Point Event Class**
- **Continuous Event Class**
- **Tour Event Class**

5.1. Point Event Class

Point Events represent point-shaped real objects that may have references to one or more routes (e.g. signpost locations). Point Events have a unique name, are managed as points within the LRS-Editor, and are additionally referenced by one or more Base Points. These Base Points lie at right angles on the referenced routes and store the measured value and the related route ID.

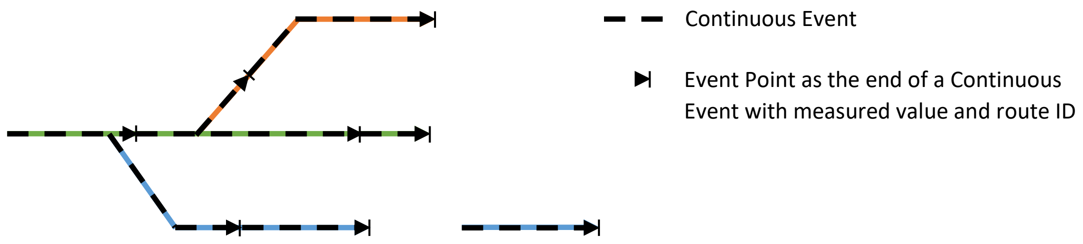


The class containing the Base Points is managed by the LRS-Editor and should not be edited manually. It is automatically added to the QGIS project as a layer.

Own custom attributes can be added to the Point Event Class and managed using QGIS.

5.2. Continuous Event Class

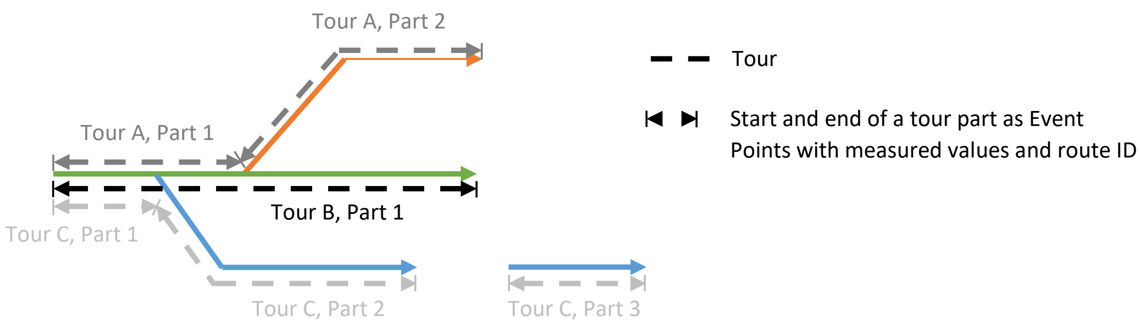
Continuous Events are linear, non-overlapping information along routes without any gaps (e.g. speed limits on roads). The Event Point at the end of a Continuous Event lies on the related route and stores an Event Name, a reference to the route as a route ID, and the measured value from the previous to the current Event Point. Event names are unique, are managed separately, and can be used or referenced by multiple event points.



Missing information along continuous features can be stored as unknown. Continuous features inherit the direction of their associated routes. Custom attributes can be added to the Event Points and managed using QGIS.

5.3. Tour Event Class

Tour Events are linear information along routes (e.g. bike tours). Unlike continuous features, they may contain gaps. A tour has a unique name, can follow different routes, and can be overlaid with one or more other tours as desired. A tour consists of one or more directed tour parts, each contains a start and an end point as Event Points on a route. A tour part stores the tour name, the belonging route ID, the measured values of the start and the end point, the direction of the tour and the position within the whole tour as a sort number. Tour parts are created, if the tour follows different routes or has a gap.



The attribute table containing information about the tour parts is managed by the LRS-Editor and should not be edited manually. It is automatically added to the QGIS project as a layer. Custom attributes can be added to a route and managed using QGIS.

6. Set up a Linear Reference Project

The following chapters, in the order presented, describe the processes of creating and managing an LRS project using the LRS-Editor.

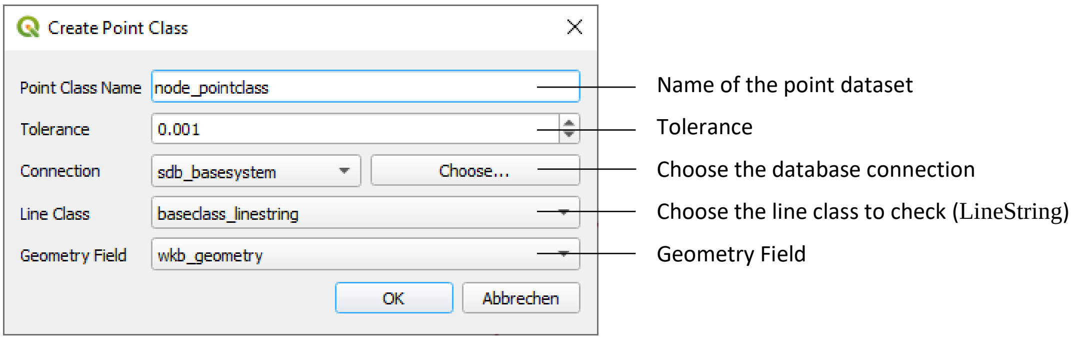
6.1. Preparations

Before creating a new LRS project, the following should be prepared:

- Complete and topologically correct line geometry and a Point Class with route definition for the Base System's settings.
- Custom topics with unique names for each object or unique event names for creating the Event Classes.

To check the line geometry of the Base System in advance, an independent point dataset can be created automatically using the menu *Create Point Class…* of the LRS-Editors; this generates the nodes of the lines.

The point dataset makes it possible to identify gaps or overlaps in the lines. The standard QGIS tools can then be used to correct these errors, thereby preparing the geometry for the Base Class.



Choose… opens the Database Settings with the database connections created in QGIS. After choosing the connection the available line datasets will be displayed. The generated point class will be stored in the same database.

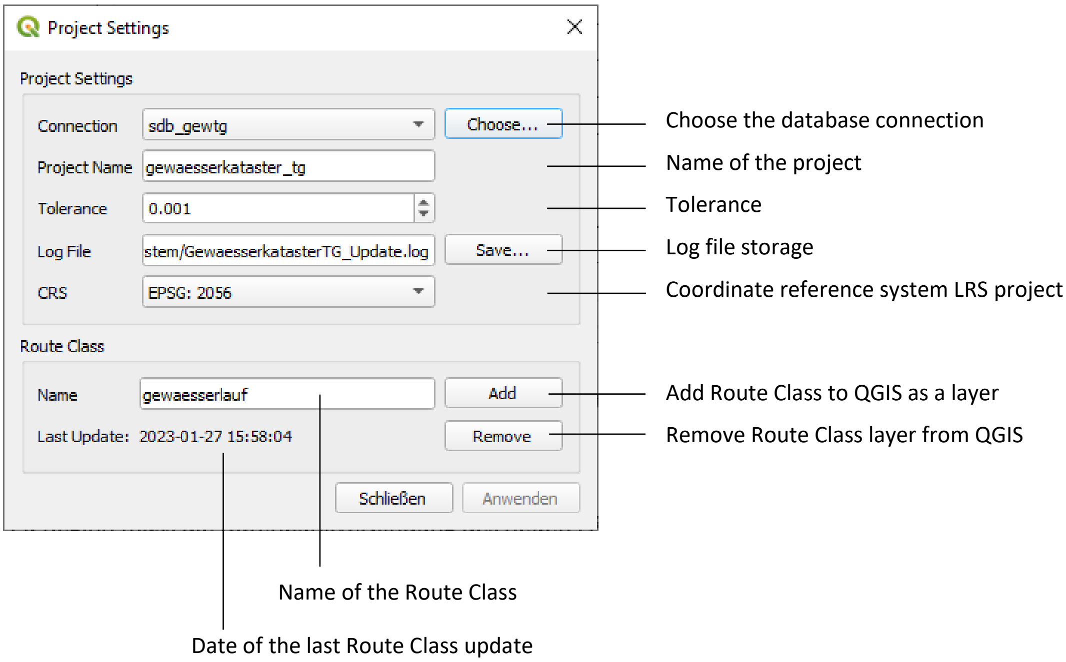
The point class contains the field pointtype with following information:

Value	Description
S	Start point of the line
E	End point of the line
P	Pseudo node: connection of two lines
T	Topology node: connection of three or more lines

If two or more points of a line are so close together that the distance between them is less than or equal to the tolerance, only a single point is generated as the centroid.

6.2. Project Setup

Under *Project Settings…* following settings are configured.



First, you need to select the database connection for the new LRS project. *Choose…* shows the *Project Database Settings* dialog with the available connections created in QGIS.

> Note: Once the dialog is closed, the QGIS project should be saved. Connection name, database name, schema and host information are not saved in the database but in the QGIS project.

The LRS project is given a unique name and a tolerance for editing the Event Points is specified.

> Note: If the distance between two Event Points is less than or equal to the tolerance value of the LRS project, the coordinates of the points are considered to be identical.

The first time, the coordinate reference system from QGIS is displayed. If you wish to select a different one, you must close the dialog – without saving – and select the desired reference system in the QGIS project properties.

The log file is created after using the update function in the toolbar, i.e. after the Base System has been checked and the Route Class updated.

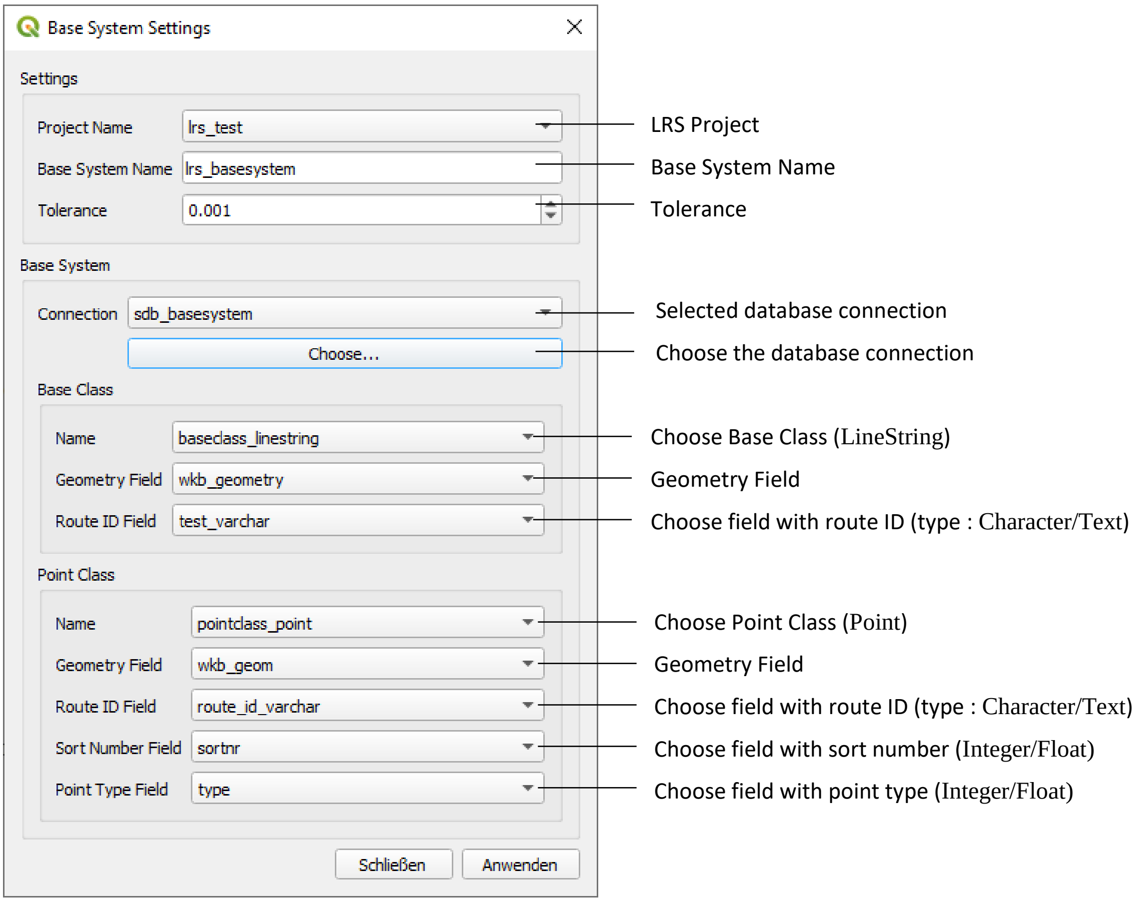
Once an update has been carried out, the date under *Last Update* is refreshed and an existing log file is replaced.

Clicking *Apply* creates the system table lrs_project with the settings and creates an empty Route Class with the desired name in the database. Once saved, the coordinate reference system and the name of the LRS project can no longer be changed.

If you reopen the project settings dialog, you can add the route class to the QGIS project as a layer.

6.3. Base System Setup

Next step includes the definition of the Base System in the dialog *Base System Settings…*.



First, you need to select the database connection for the Base System. *Choose…* shows the *Project Database Settings* dialog with the available connections created in QGIS.

> Note: Once the dialog is closed, the QGIS project should be saved. Connection name, database name, schema and host information are not saved in the database but in the QGIS project.

The name of the LRS project is automatically displayed based on the project settings. A name and a tolerance are assigned to the Base System; the tolerance is used when checking the topology of the Base System.

> Note: If the distance between two points of the Point Class is less than or equal to the tolerance value, the coordinates of the points are considered to be identical.

The tables in the database are displayed in the drop-down lists for selecting the Base and Point class, taking the geometry type into account. Once both classes have been specified, drop-down lists below display the available fields for both classes.

For the route ID field type is Character/Text, for the sort number and point type Float oder Integer values are expected.

The field point type accepts following values:

Value	Description
1	Start of the route or the route part
2	End of the route or the route part

> Note: The sort number indicates the correct position of the route part within the entire route. It always starts with the number 1. If new route parts are added or existing ones deleted, the numbers of the following parts of the route must be adjusted accordingly.

The lines in the Base Class may only be of the type LineString (Basisklasse), no circular or arc types are supported. In the Base and the Point Class no multipart features are allowed.

Once the information has been saved, the system table lrs_basesystem is created in the database.

6.3.1. Loop route

Loop routes are supported under the following conditions:

- The starting point or vertex of the first line matches the last one of the last line of the loop in the Base Class.
- The points in the Point Class associated with the loop route are identical in position, i.e. they lie on top of one another.
- Base and Point class have the field id, which has a unique identifier, e.g. an integer value.
- The loop route must not consist of several linked route parts; the route ID for loop routes must not occur more than once.

The order of the lines associated with the loop route is determined by the ascending value of the id field. The direction of the lines is taken from the Base Class; in other words, the direction of the Base Class lines determines the orientation of the route.

Complex composite routes, such as lasso routes, are not supported or must be created from different route IDs.

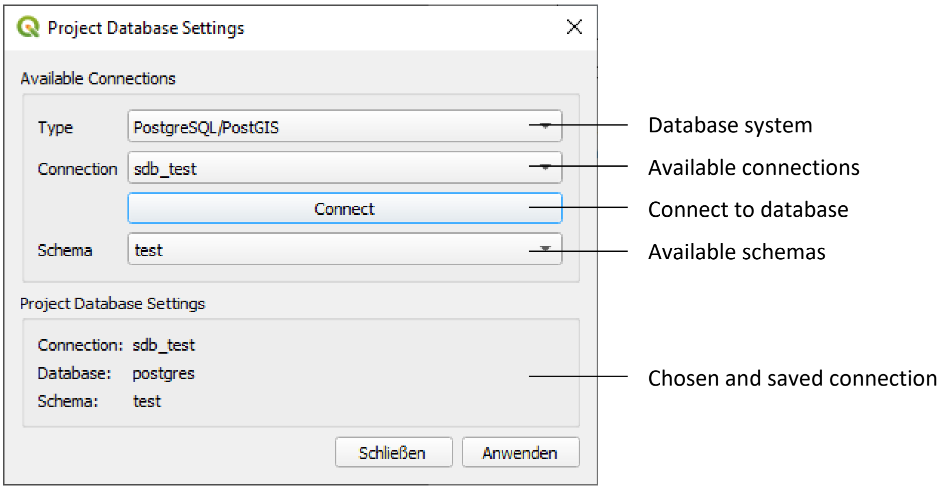
6.4. Database Settings

The dialog *Database Settings* dialog is accessed via a previously opened dialog and allows you to select an available database connection.

The connections are previously created in QGIS, and the login details are stored in the QGIS authentication database.

> Note: The LRS-Editor does not store or manage any login details; instead, it always accesses QGIS's settings and authentication database to establish the connection.

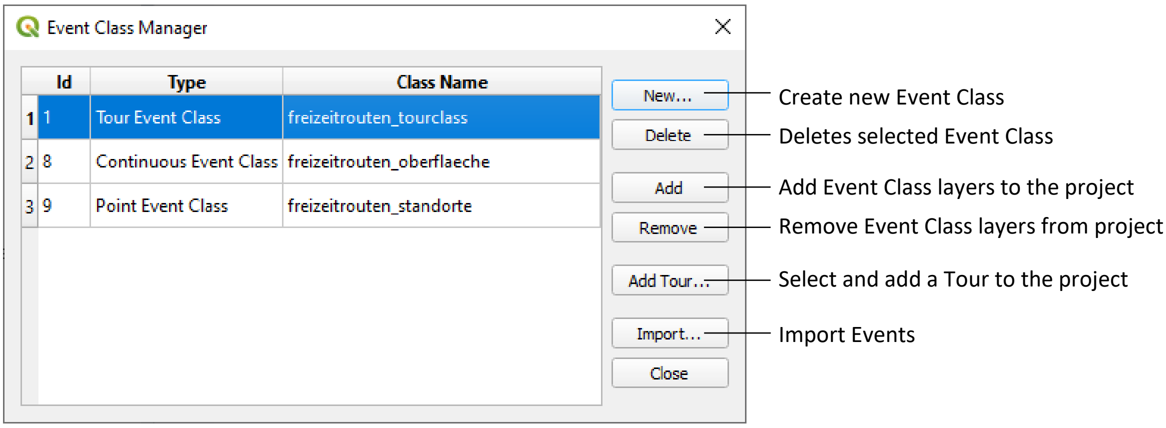
The dialog displays all PostgreSQL connections created with QGIS. Clicking the *Connect* button checks the connection and displays the available database schemas in the list below. Clicking *Apply* selects the connection with the chosen schema, and the details appear in the lower section of the dialog.



When the dialog is closed, the selected database connection is displayed in the previously opened dialog.

6.5. Event Class Setup

With the dialog *Event Class Manager…* Event Classes can be created and managed.



Clicking *New…* opens the dialog to choose and create the Event Class type. The dialog requires a unique Event Class name, which cannot be changed once the class has been created.

With *Add* the layers according chapter 7 of the Event Class will be added to the QGIS Project. With *Remove* the specific layers will be removed from project, but not deleted in the database.

> Note: Layers belonging to an event class should only be added to or removed from the QGIS project via the Event Class Manager dialog.

Using *Add Tour…*, you can select individual tours from a Tour Event Class and add them to the QGIS project as separate layers, or remove them again. A tour's layer is created as a view in the database; it is read-only and is automatically updated when changes are saved during editing the related tour. The *Add Tour…* button is only active if a Tour Event Class is selected in the left table.

Event Classes can only be completely and correctly deleted from the database using *Delete* if the associated layers have first been removed from the QGIS project using *Remove*. This also applies to the layers of individual tours of a Tour Event Class that have been added to the QGIS project using *Add Tour…*.

The *Import*… function for importing Events in an existing Event Class will be explained in chapter Data Import.

Information of the *Event Class Manager* will be saved in system table lrs_event_classes.

7. Editing Event Classes

Following layers are in QGIS created, when adding the Event Classes to the project:

Type	Layer	Editing	Undo
Point Event Class	Point Event	Yes	Yes
	Base Points (suffix _bp)	No	Yes
Continuous Event Class	Event Points	Yes	Yes
	View with line geometry (prefix v_)	No	No
Tour Event Class	Event Points	Yes	Yes
	Table with tour parts (suffix _mt)	No	Yes
	View with line geometry (prefix v_)	No	No

The views are the linear representation of the measured values along the route and are read-only. They are automatically updated when changes are saved during editing.

Editing with the LRS-Editor toolbar is activated, when the edit status in QGIS is turned on. To start, the mentioned layer above must be selected before. The other layers of the Event Classes are automatically updated in edit mode; their data or features should not be edited manually.

> Warning: LRS tables or features should not be edited or modified using other tools in QGIS or in other systems. This may lead to inconsistencies in the datasets and cause the LRS-Editor to malfunction.

As long as a layer of an Event Class is in edit mode and a tool from the toolbar is active, individual edits in QGIS can be undone via *Edit* > *Undo*. Once changes to Event Classes have been saved to the database, this is no longer possible.

> Warning: If changes are undone, the changes to the associated layers (base points _bp and attribute table _mt) must also be undone, as QGIS does not support undoing multiple layers at the same time.

Edits are automatically saved as well, whenever the active LRS tool is deactivated – for example, when you click on a different tool.

It is recommended to work with the snapping enabled.

If another layer, not belonging to the LRS project, is turned on for editing, the tools of the LRS-Editor toolbar are disabled.

Manage additional custom fields

Following layers allow the administration of own custom fields:

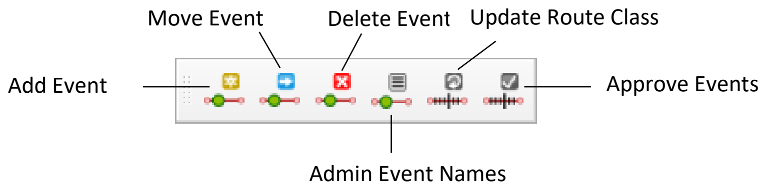
Type	Layer
Point Event Class	Point Event
Continuous Event Class	Event Points
Tour Event Class	Table with tour parts (suffix _mt)

> Warning
>
> When managing custom attributes, care must be taken not to modify any of the LRS fields or values of an event class. Doing so may lead to inconsistencies in the data records and cause the LRS-Editor to malfunction.
>
> For custom fields, the storage of NULL values should be permitted. Otherwise, for example when adding Event Points, the LRS Editor will be unable to write them correctly to the database. This also leads to inconsistencies in the data records.

The custom fields of a Continuous or Tour Event Class can also be displayed in the view by refreshing the relevant view after adding the custom fields via the *Event Class Manager*.

If the Shift key is held down at the same time, the functions described in chapter 6.5 for creating (*New*…), deleting (*Delete*), adding (*Add*) and removing (*Remove*) an event class apply only to the view. After adding custom fields, the associated view can therefore be removed from the project, deleted from the database, recreated and added back to the project – using simultaneously the Shift key. If a custom field has to be removed again, the corresponding view must first be removed from the project and deleted. The custom field can then be removed from the layer. Once the view has been recreated, it will no longer hold the field.

7.1. Toolbar Overview



The LRS-Editor toolbar provides following tools. The 'Activated' column indicates when a tool is enabled and can therefore be used:

Tool	Description	Activated
Add Event	Creating a new Point Event or Event Point	Edit mode is turned on Layer is selected
Move Event	Move an existing Point Event or Event Point	Edit mode is turned on Layer is selected
Delete Event	Delete an existing Point Event, Event Point or Base Point	Edit mode is turned on Layer is selected
Admin Event Names	Manage the Event Names	Layer is selected Edit mode is turned off
Update Route Class	Check Base System, update Route Class and Event Classes	Edit mode is turned off
Approve Events	Approve changed Event Points or Base Points after Update	Edit mode is turned off

You can show or hide the toolbar by right-clicking on the QGIS toolbar.

Contains an Event Class not approved Events, the tools cannot be used. A dialog will be displayed prompting you to approve the pending Events first.

7.2. Add Event

New Event Names can also be added in advance via the Event Name Manager dialog (tool Admin Event Names).

If during digitalization more than one route is selected – in case the lines of the routes are close together – the desired route can be selected in the appearing dialog.

7.2.1. Point Event Class

1. Click on the route on the map that you want to reference the new Event to. Hold down the Shift key to select additional routes, or press Ctrl to deselect routes. At least one route must be selected.
2. Click on the location of the new object on the map.
3. Enter a new Event Name in the dialog. The name will be checked for uniqueness. If there are unreferenced Event Names in the Event Class, the name of the new object can be selected from a list. If you *Cancel* the list without selecting, you can then enter a new name instead afterwards.
4. A new Point Event and the associated Base Point will be inserted.

7.2.2. Continuous Event Class

1. First, select an Event Name from the QGIS dock widget on the left. You can filter the available Event Names by entering the name in the text field.
2. Clicking on the desired route on the map sets the end point – in the direction of the route – of the Continuous Event. If there is no Event on the route yet, the Event Point is automatically set at the end of the route.
3. The new Event Point will be inserted. A message is displayed if the event points have the same Event Name above or below along the route. Saving the edits writes the point to the database and updates the corresponding view.

7.2.3. Tour Event Class

1. The first click on the desired route determines the start point of the tour or the first tour part. An empty square indicates the starting point.
2. Give the name of the new tour in the following dialog. The name will be checked for uniqueness. If there are unreferenced tours in the Event Class, the name of the new tour can be selected from a list. If you *Cancel* the list without selecting, you can then enter a new name instead afterwards.
3. Clicking on the same route a second time determines the end of the tour or the end of the first tour part, and thus the direction of the tour. If a different route is selected, the square at the starting point remains in place until the end point on the same route is digitised or the tool is exited. If you hold down the Shift key while digitising end point, you can add further tour parts of the tour. Adding tour parts follows the same procedure, without prompting for the tour name. The tour is finished once you release the Shift key at the end point of the final tour part.
4. Once the tour or the tour part has been completed, the start and end points will be inserted. Saving the edits writes the point to the database and updates the corresponding view.

7.3. Move Event

If during digitalization more than one route is selected – in case the lines of the routes are close together – the desired route can be selected in the appearing dialog.

7.3.1. Point Event Class

1. Select the Point Event you want to move by clicking on the object. The associated Base Points will be selected.
2. Clicking on the map a second time sets the new position of the Point Event. The LRS-Editor automatically updates the positions of the associated Base Points. If you click on a route while holding down the Shift key, an additional Base Point of the Point Event will be inserted on the route.

7.3.2. Continuous Event Class

1. Select the Event you want to edit by clicking on the corresponding Event Point. If there are several Event Points close together, you can select the desired Event in the following dialog. The corresponding Event Name is displayed in the dock widget on the left.
2. Clicking on the same route a second time sets the new position respectively the new end of the Event. Event Points cannot be moved beyond a neighbouring Event Point or to the beginning of a route. Event Points at the end of a route cannot be moved.
3. Saving the edits writes the point to the database and updates the corresponding view.

Instead of moving the Event, you can change the Event Name in the dock widget on the left after selecting the Event Point. A message is displayed if the event points have the same Event Name above or below along the route.

7.3.3. Tour Event Class – Edit Tour Part

1. Select the start or end of the tour part you want to adjust by clicking on the corresponding Event Point. If there are several Event Points close together, you can select the desired tour part in the dialog.
2. Clicking on the same route a second time sets the new start or end of the tour or tour part. Event Points of tour parts cannot be moved over tour parts of the same tour.
3. Saving the edits writes the point to the database and updates the corresponding view.

7.3.4. Tour Event Class – Add Tour Part

1. Select the beginning or end of an existing tour part – depending on whether you want to insert the new part before or after the existing one – by clicking on the corresponding Event Point. If there are several Event Points close together, you can select the desired tour part in the dialog.
2. By clicking once while holding down the Shift key, you can set the starting point of the new tour part either on the same route or on a new one. An empty square indicates the position of the starting point.
3. Clicking a second time on the same route as the first point while holding down the Shift key defines the end of the tour part. If a different route is selected, the square at the starting point remains until the end point is marked on the same route while holding down the Shift key or until the tool is exited. It is not possible to add a tour part in the opposite direction of an existing tour part of the same tour on the route. The system also checks whether tour parts of the same tour overlap on the same route.
4. Saving the edits writes the point to the database and updates the corresponding view. The sort numbers for the following tour parts of the same tour are automatically adjusted.

7.4. Delete Event

If multiple Point Events or Event Points are selected during deletion and they are located close to each other, you can select the desired point in a dialog.

Removing an Event does not delete the associated Event Name. The Event Name can still be used and appears in the list dialogs. It can be permanently removed from the database using the Event Name Manager dialog (tool Admin Event Names).

7.4.1. Point Event Class

1. Clicking on a Point Event deletes it with its associated Base Points. Instead of the Point Event, you can also delete individual Base Points.

7.4.2. Continuous Event Class

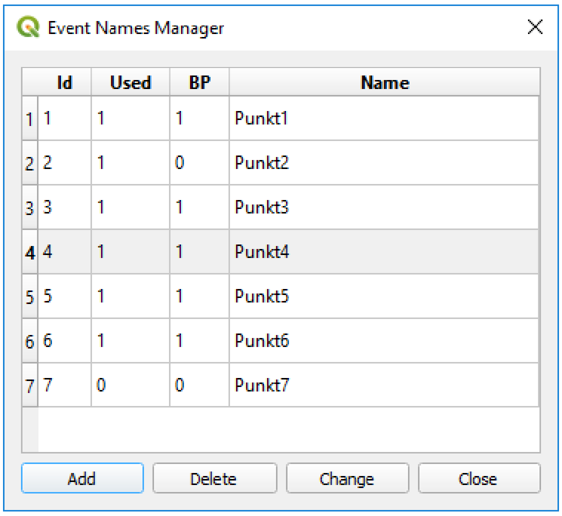
1. Clicking on an Event Point deletes the Continuous Event. A message is displayed if the event points have the same Event Name above or below along the route. Event Points at the end of a route can only be deleted once all Continuous Events along the route have already been removed.

7.4.3. Tour Event Class

1. Clicking on an Event Point deletes that part of the tour. If you hold down the Ctrl key while clicking, the entire tour – including its name – will be permanently deleted from the database.

7.5. Admin Event Names

The function opens the Event Names Manager (if the selected layer of the Event Class is not in edit mode). With the dialog new Event Names can be added, changed or deleted.



The *Used* column shows the number of references of the Event Name. For Point Event classes, the additional *BP* column shows the number of Base Points for the Point Event. Clicking on a column header sorts that column.

When creating new Event Names or modifying existing ones, the system checks that the Event Name is unique. Event Names can only be deleted if they are no longer in use, i.e. if they have no remaining references.

> Note: Changes made in the Event Names Manager are saved directly to the database and cannot be undone.

7.6. Update Route Class

The function updates the Route Class of the LRS project and is executed by the user when:

- A Base System has been newly set up and the Route Class is to be generated for the first time
- Changes have been made to the Base System, i.e. to the Base or Point Class, by the user

The function checks first whether:

- All layers of an Event Class are present in the QGIS project
- No layer of an Event Class is in edit mode
- A layer contains not approved Events

In these cases, the process is not started.

Following steps are performed:

1. **Check Base System**: Checking the topology of the Base Class with its line geometry and the routes, defined in the Base and Point Class All routes are generated and stored in the internal class lrs_route_class.
2. **Update Route Class**: The routes generated in lrs_route_class are compared with those in the Route Class of the LRS project. Only valid routes from lrs_route_class are taken into account. For routes to be deleted in the Route Class, a check is carried out to see if any Events are referenced with the route. If so, the route is not deleted. New routes are added to the route class, whilst those with geometric changes are passed on for updating the associated Events.
3. **Update Event Classes**: Events that lie directly along the modified line geometry are repositioned if the distance between the old and new positions of the Event Point is greater than the tolerance specified in the project settings. In this case, the measured value is recalculated. In addition, the date in the field apprtstz is set to Null. Those Events must be verified using the function *Approve Events*.

   Events above changed route geometry get updated measured values, azimuth and a new date in the field changetstz. Events below changed route geometry remain unchanged.

Results are written to the log file. An explanation of the messages can be found in the appendix. Once the process has been completed, the date is updated in the project settings.

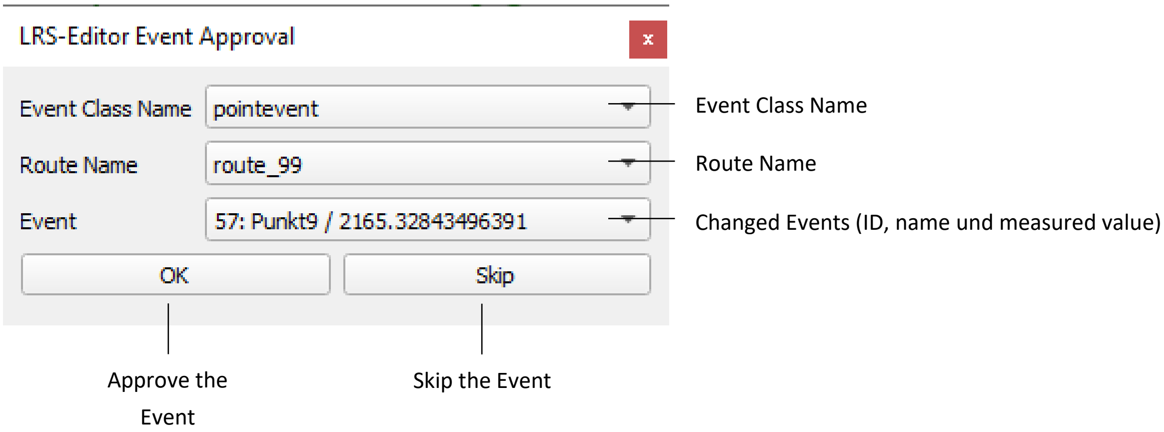
If routes are to be deleted that have associated Events, a dialog at the end of the update process shows the affected routes and Event Classes. The dialog allows the routes and all referenced Events to be deleted automatically. That means all Event Points, Point Events and its Base Points will be removed of a deleted route. Event Names of the Event Classes however will not be removed in that process.

7.7. Approve Events

The function checks first whether:

- All layers of an Event Class are present in the QGIS project
- No layer of an Event Class is in edit mode

If there are after a Route Update changed Events, the widget Event Approval must be opened, before the edit tools of the toolbar can be used:



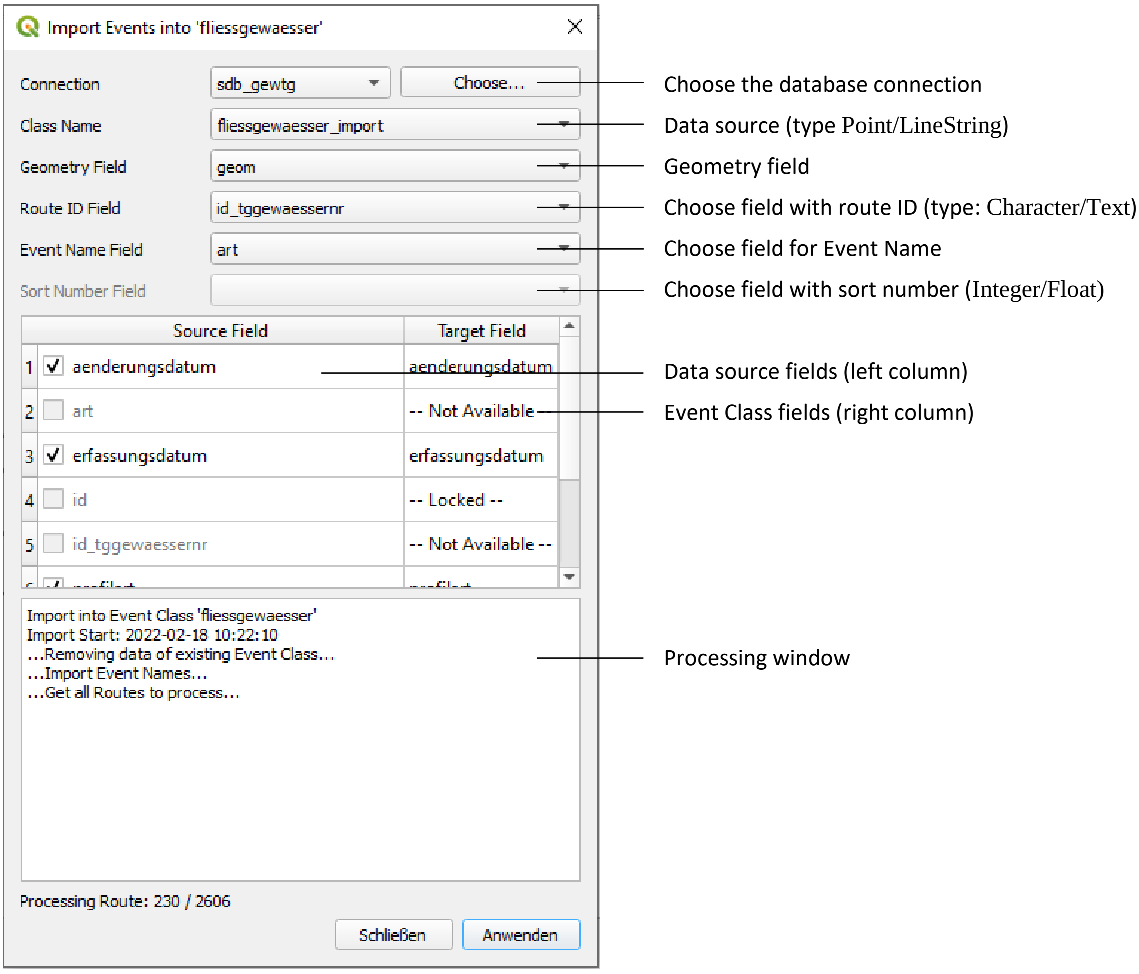
All Events are listed for which the date in the field apprtstz was set to Null. They must now be approved. The map view is moved to the active Event of the list and the corresponding Event Point is selected. Clicking OK updates the date, removes the event from the list and displays the next one. Clicking Skip means the event is not confirmed, is skipped and the view moves to the next event in the list.

Once all pending Events have been confirmed, the widget can no longer be opened and all other tools on the toolbar can be used again.

If the confirmation relates to a Continuous or Tour Event Class, the measured values of the Events along the selected route is also checked. Incorrect measures (overlaps or gaps) is displayed in a text field below on the widget.

8. Data Import

Existing gis data stored in a PostgreSQL database can be imported as a data source into an empty Event Class, which must already exist in the QGIS project. To do this, first select the Event Class into which the data is to be imported in the *Event Class Manager* dialog, and then open the following dialog using the *Import* function.



First, you need to select the database connection for the Base System. *Choose…* shows the Project *Database Settings* dialog with the available connections created in QGIS.

The data sources from the selected database are displayed based on the geometry type. For a new Point Event Class Points and for Continuous and Tour Event Class LineString can be imported.

For all types of Event Classes following additional information is required:

- **Geometry Field**: The coordinate reference system of the data source must match that of the empty Event Class.
- **Route ID Field**: This field contains the name of the route by which the object to be imported should be referenced. If the route name does not exist in the LRS project's Route Class, the object will not be imported. If a point to be imported into a Point Event Class has references to multiple routes, it must first be entered multiple times in the data source with different route names.
- **Event Name Field**: Contains the name of the Event to be imported.

- **Sort Number Field**: The list box is enabled only for Tour Event Classes. It allows you to specify a field containing the sort number for multi-part tours. The sort number indicates the position of a tour part within the entire tour. If <None> is selected, each object or line to be imported is interpreted as a separate tour, and the sort number is set to 1.

The data in in the specified fiels must not contain Null values. The import will abort, if Null values are present.

If the Event Class selected for the import is not empty, the associated tables can be cleared after startup automatically.

The tolerance specified in the *Project Settings* is applied when creating Event Points during the import.

If the lines to be imported into a **Continuous Event Class** do not fully cover a route in the LRS project's Route Class, the Event Name 'n/a' is automatically assigned to the remaining sections of that route. At the end of the import, the value 'n/a' can also be assigned to routes that do not have any Continuous Event.

Points to be imported into a **Point Event Class** must have a unique name. Points to be imported that have the same name and refer to the same route will be skipped.

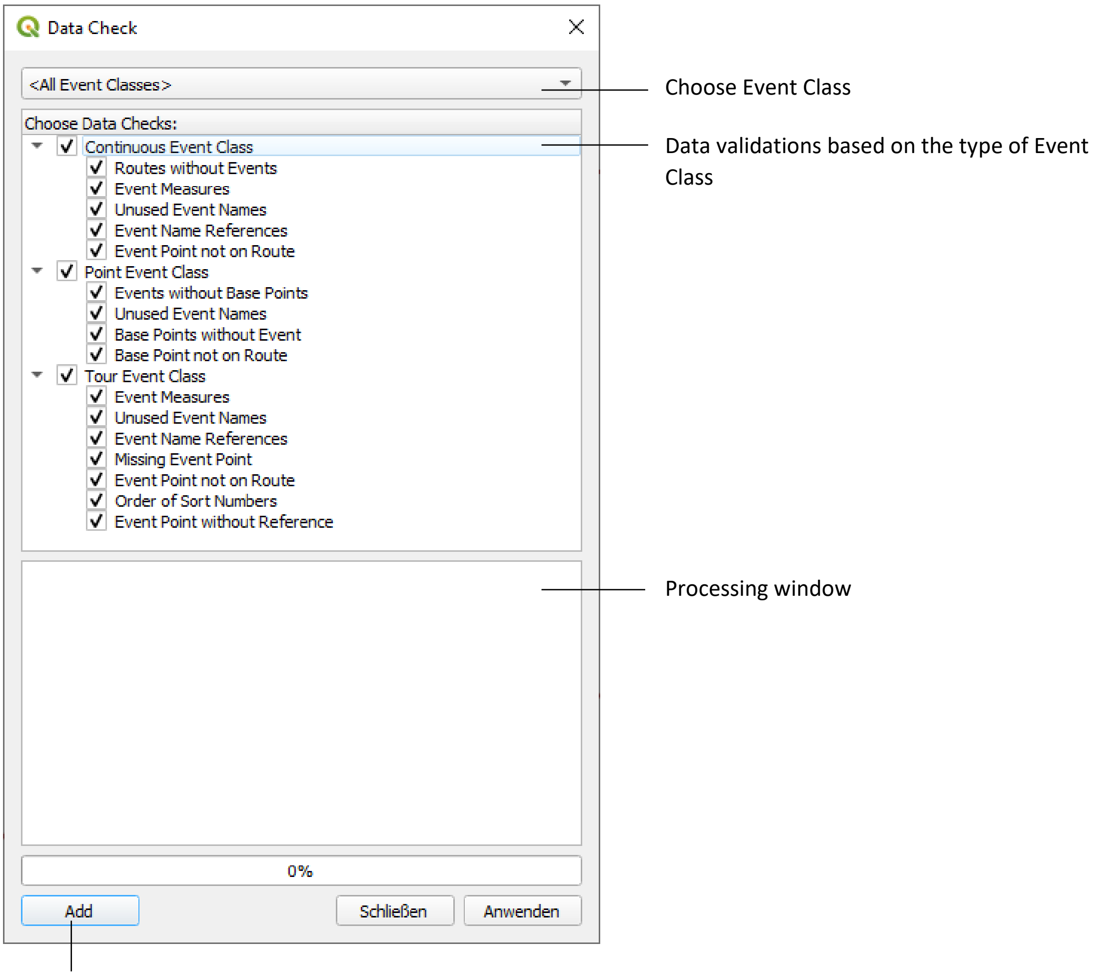
It is recommended to perform a data check after the import is complete, especially for Continuous Event Classes. This can help identify overlaps or gaps in the data source.

Import custom values

This dialog allows you to import custom values stored in additional fields of the data source (*Source Field*). To import these values, the target fields must first be manually created in the empty Event Class (*Target Field*), and the field names must match those in the data source. The type of the data source fields must match the type of the target fields; otherwise, the import will fail. Reserved system fields of the empty Event Class are displayed as disabled (*Locked*).

9. Data Check

Using *Data Check…* you can validate the data of the Event Classes and save the results in a separate class named lrs_check_class.



Resultate als Layer lrs_check_class dem Projekt hinzufügen

The following tests are performed depending on the selected Event Classes:

Continuous Event Class

- *Routes without Events*: Route names that do not have any Continuous Events.
- *Event Measures*: Check whether Continuous Events along a route overlap or have gaps – faulty Event Points are displayed.
- *Unused Event Names*: Event Names that have not been used yet.
- *Event Name References*: Exports the UUIDs of Event Names that cannot be found in the Event Name table - along with the corresponding Event Points.
- *Event Point not on Route*: Export of Event Points that do not lie on the referenced route, taking the project tolerance into account.

Point Event Class

- *Events without Base Points*: Export of Point Events without corresponding Base Points.
- *Unused Event Names*: Event Names that have not been used yet.
- *Base Points without Event*: Export of Base Points with missing Point Event.
- *Base Point not on Route*: Export of Base Points that do not lie on the referenced route, taking the project tolerance into account.

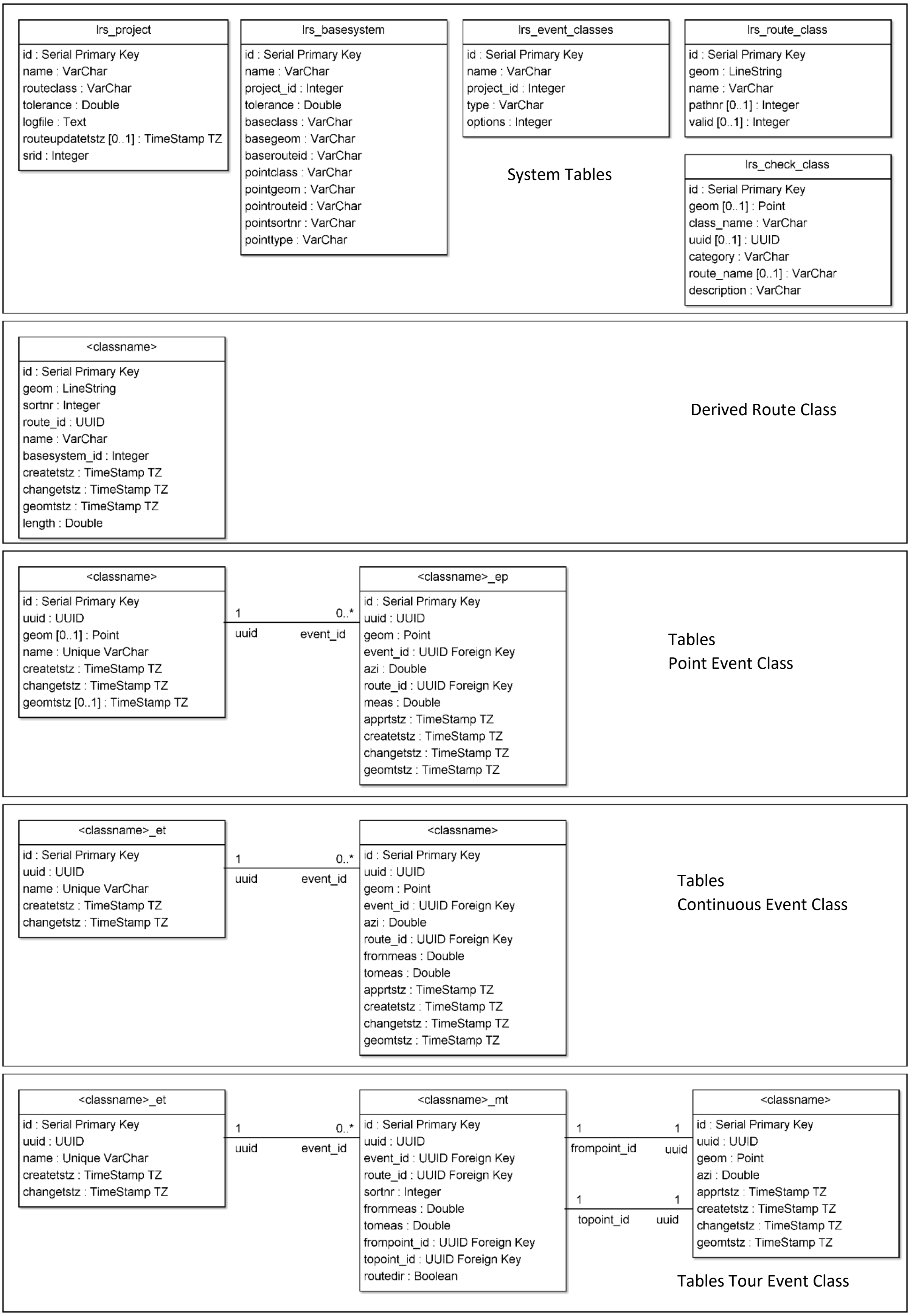
Tour Event Class

- *Event Measures*: Check whether tour parts of the same tour overlap on the same route. Exports faulty Event Points.
- *Unused Event Names*: Tour Names that have not been used yet.
- *Event Name References*: Exports the UUIDs of Event Names that cannot be found in the Event Name table.
- *Missing Event Point*: Exports the UUIDS of the Event Points – based on the table with the tour parts – that cannot be found in the table with the Event Points.
- *Event Point not on Route*: Export of Event Points that do not lie on the referenced route, taking the project tolerance into account.
- *Order of Sort Numbers*: Checks the sort number of the tour parts of every tour.
- *Event Point without Reference*: Export of the Event Points without reference in the table with the tour parts.

Before each data check the system table lrs_check_class will be cleared.

10. Appendix

10.1. Data Model



10.1.1. System Tables

lrs_project			
Field	Data Type	Cardinality	Description
id	Integer	1	Serial Primary Key
name	VarChar(100)	1	Project Name
routeclass	VarChar(100)	1	Route Class Name
tolerance	Double	1	Project Tolerance for Event Classes must be greater than 0
logfile	Text	1	Path to the log file
routeupdatetstz	TSTZ	0..1	Timestamp with Timezone Date of the last Route Class Update
srid	Integer	1	SRID Spatial Reference System Identifier

Note: The Spatial Reference System Identifier (SRID) should not be changed in the database after the LRS project has been set up for the first time.

lrs_basesystem			
Field	Data Type	Cardinality	Description
id	Integer	1	Serial Primary Key
name	VarChar(100)	1	Base System Name
project_id	Integer	1	Foreign key Reference to the ID of the project
tolerance	Double	1	Base System Tolerance must be greater than 0
baseclass	VarChar(100)	1	Base Class name
baserouteid	VarChar(100)	1	Field name route ID of the Base Class
basegeom	VarChar(100)	1	Geometry field name of the Base Class
pointclass	VarChar(100)	1	Point Class name
pointgeom	VarChar(100)	1	Geometry field name of the Point Class
pointrouteid	VarChar(100)	1	Field name route ID of the Point Class
pointsortnr	VarChar(100)	1	Field name route sort number of the Point Class
pointtype	VarChar(100)	1	Field name with point type Allowed values: 1 = Start of the route or the route part, 2 = End of the route or the route part

lrs_event_classes			
Field	Data Type	Cardinality	Description
id	Integer	1	Serial Primary Key
name	VarChar(100)	1	Name of the Event Class
project_id	Integer	1	Foreign key Reference to the ID of the project
type	VarChar(2)	1	Event Class type: c = Continuous Event Class p = Point Event Class t = Tour Event Class
options	Integer	1	Event Class options Not in use

lrs_route_class			
Field	Data Type	Cardinality	Description
id	Integer	1	Serial Primary Key
geom	LineString	1	Geometry
name	VarChar(100)	1	Route name
pathnr	Integer	0..1	Part index of multi-part routes Default = 1
valid	Integer	0..1	Route status; default = 1 1 = valid route 0 = route with errors -> no Route Class Update

Internal route class with all routes for Route Class comparison, created each time when Base System is checked using the Update function.

lrs_check_class			
Field	Data Type	Cardinality	Description
id	Integer	1	Serial Primary Key
geom	Point	0..1	Geometry
class_name	VarChar(100)	1	Event Class name
uuid	UUID	0..1	Universally Unique Identifier, Version 4 UUID of the Event Point
category	VarChar(10)	1	Type: ERROR, WARNING, INFO
route_name	VarChar(100)	0..1	Route name
description	VarChar(200)	1	Description

Resulting table after data checks.

10.1.2. Route Class

<classname>			
Field	Data Type	Cardinality	Description
id	Integer	1	Serial Primary Key
geom	LineString	1	Geometry
sortnr	Integer	1	Sort number of multi-part routes, begins with 1
route_id	UUID	1	Universally Unique Identifier, Version 4
name	VarChar(100)	1	Route name
basesystem_id	Integer	1	Foreign key Reference to the ID of the Base System
createtstz	TSTZ	1	Timestamp with Timezone Date of creation
changetstz	TSTZ	1	Timestamp with Timezone Date of change
geomtstz	TSTZ	1	Timestamp with Timezone Date of geometry changes
length	Double	1	Route length

Derived Route Class of the LRS project containing only valid routes, generated from the Base System for the measured referencing of Event Classes. Class name according lrs_project.routeclass. The combined values of name und sortnr build together a unique key as well.

10.1.3. Point Event Class

<classname>			
Field	Data Type	Cardinality	Description
id	Integer	1	Serial Primary Key
uuid	UUID	1	Universally Unique Identifier, Version 4
geom	Point	0..1	Geometry
name	VarChar(100)	1	Unique Name of the Point Event (Unique Constraint)
createtstz	TSTZ	1	Timestamp with Timezone Date of creation
changetstz	TSTZ	1	Timestamp with Timezone Date of change
geomtstz	TSTZ	0..1	Timestamp with Timezone Date of geometry changes

Class with Point Events

<classname>_ep			
Field	Data Type	Cardinality	Description
id	Integer	1	Serial Primary Key
uuid	UUID	1	Universally Unique Identifier, Version 4
geom	Point	1	Geometry
event_id	UUID	1	Foreign key Reference to Point Event
azi	Double	1	Azimuth of the Base Point along the route
route_id	UUID	1	Foreign key Reference to route ID
meas	Double	1	Measured value along route
apprtstz	TSTZ	1	Timestamp with Timezone Approval date
createtstz	TSTZ	1	Timestamp with Timezone Date of creation
changetstz	TSTZ	1	Timestamp with Timezone Date of change
geomtstz	TSTZ	1	Timestamp with Timezone Date of geometry changes

Automatically updated class containing the Base Points of a Point Event

10.1.4. Continuous Event Class

<classname>_et			
Field	Data Type	Cardinality	Description
id	Integer	1	Serial Primary Key
uuid	UUID	1	Universally Unique Identifier, Version 4
name	VarChar(100)	1	Unique name of the Continuous Event (Unique Constraint)
createtstz	TSTZ	1	Timestamp with Timezone Date of creation
changetstz	TSTZ	1	Timestamp with Timezone Date of change

Table for managing the Event Names; not visible in QGIS.

The corresponding view of a Continuous Event Class is composed of fields from both tables. Therefore, no description of the view is provided here.

<classname>			
Field	Data Type	Cardinality	Description
id	Integer	1	Serial Primary Key
uuid	UUID	1	Universally Unique Identifier, Version 4
geom	Point	1	Geometry
event_id	UUID	1	Foreign key Reference to a Continuous Event
azi	Double	1	Azimuth of the Event Point along the route
route_id	UUID	1	Foreign key Reference to route ID
frommeas	Double	1	From measure along route
tomeas	Double	1	To measure along route
apprtstz	TSTZ	1	Timestamp with Timezone Approval date
createtstz	TSTZ	1	Timestamp with Timezone Date of creation
changetstz	TSTZ	1	Timestamp with Timezone Date of change
geomtstz	TSTZ	1	Timestamp with Timezone Date of geometry changes

Class with Event Points.

10.1.5. Tour Event Class

<classname>_et			
Field	Data Type	Cardinality	Description
id	Integer	1	Serial Primary Key
uuid	UUID	1	Universally Unique Identifier, Version 4
name	VarChar(100)	1	Unique name of a tour (Unique Constraint)
createtstz	TSTZ	1	Timestamp with Timezone Date of creation
changetstz	TSTZ	1	Timestamp with Timezone Date of change

Table for managing the Event Names; not visible in QGIS.

<classname>			
Field	Data Type	Cardinality	Description
id	Integer	1	Serial Primary Key
uuid	UUID	1	Universally Unique Identifier, Version 4
geom	Point	1	Geometry
azi	Double	1	Azimuth of the Event Point along the route
apprtstz	TSTZ	1	Timestamp with Timezone Approval date
createtstz	TSTZ	1	Timestamp with Timezone Date of creation
changetstz	TSTZ	1	Timestamp with Timezone Date of change
geomtstz	TSTZ	1	Timestamp with Timezone Date of geometry changes

Class with Event Points.

<classname>_mt			
Field	Data Type	Cardinality	Description
id	Integer	1	Serial Primary Key
uuid	UUID	1	Universally Unique Identifier, Version 4
event_id	UUID	1	Foreign key Reference to the tour name
route_id	UUID	1	Foreign key Reference to route ID
sortnr	Integer	1	Sort number of the tour part, starting with 1
frommeas	Double	1	From measure along route
tomeas	Double	1	To measure along route
frompoint_id	UUID	1	Foreign key Reference to Event Point
topoint_id	UUID	1	Foreign key Reference to Event Point
routedir	Boolean	1	Direction of the tour part True = in direction of the route False = not in direction of the route

Table with tour parts.

The corresponding view of a Tour Event Class is composed of fields from all tables. Therefore, no description of the view is provided here.

10.2. Updated Date Fields

Field	Description
apprtstz	Available only at Event Points Null for not approved, changed events; updated upon approval
createtstz	Create date of an object
changetstz	Date on which a field of the object was modified
geomtstz	Date on which an object's geometry was modified

When a new object is created, the current date is entered into all four of the object's fields.

10.3. Log File

The Log File contains the results of the Function Route Class Update. They are written to the file in the order described there. All messages are listed here for reference.

Base System Check: CHECK DATA OF BASE SYSTEM

Type	Message	Description
ERROR	Base Class: Field <Field Name> has NULL values	Field <Field Name> of the Base Class contains Null values
ERROR	Point Class: Field <Field Name> has NULL values	Field <Field Name> of the Point Class contains Null values

If errors are found during data check, the function will terminate.

Base System Topology Check: CHECK TOPOLOGY OF BASE SYSTEM

Type	Message	Description
ERROR	Route <Route Name>: Number of paths does not match number of points, route not valid	The number of paths or route parts, consisting of the corresponding lines in the Base System, does not match the number of corresponding points in the Point Class. This may be due to gaps between the start and end points of lines that belong together.
ERROR	Route <Route Name>, Path <Route Part Number>: No Points found, route not valid	No points in the Point Class were found that correspond to the route
ERROR	Route <Route Name>: SortNr of points do not match, route not valid	The sort numbers of the points associated with the route are incorrect
ERROR	Route <Route Name>, SortNr <Sort Number>: Start- or endpoint not in tolerated distance, route not valid	The start or end point is not within the allowed tolerance from the line
ERROR	Route <Route Name>, SortNr <Sort Number>: Start- and endpoint are of same type, route not valid	The start and end points are of the same type
ERROR	Route <Route Name>, SortNr <Sort Number>: Startpoint is not of correct type, route not valid	The start point of the route does not have the correct point type

ERROR	Route <Route Name>, SortNr <Sort Number>: Endpoint is not of correct type, route not valid	The end point of the route does not have the correct point type
ERROR	Route <Route Name>: Points without LineString	Start or end points without an associated line in the Base Class
INFORM	Route <Route Name>, Path <Route Part Number>: Path-Nr of route will be changed	The route part number is adjusted to match the sort number
INFORM	Route <Route Name>, SortNr <Sort Number>: Start- and endpoint have identical positions	The start and the end point are on top of each other; this may be a loop route
INFORM	Route <Route Name>, Path <Route Part Number>: LineString in reversed direction	The Base Class line(s) of the route are in the opposite direction to the route

If errors are detected during the topology check, the corresponding route will not be created or updated.

Update Route Class: UPDATE ROUTE CLASS

Type	Message	Description
ERROR	Route <Route Name>: Basesystem does not exist anymore, but still has events of class <Event Class Name>	The route no longer exists in the Base System, but still has Event Points belonging to an Event Class
ERROR	Route <Route Name>: coordinates are not in the same order. Route not updated.	The coordinates of the corresponding lines in the Base System are not in the correct order. The route is not being updated. The direction of the route (or parts of it) may have been changed.
INFORM	Route <Route Name>: new route inserted	New route created and inserted into Route Class
INFORM	Route <Route Name>: deleted	Route deleted in the Route Class
INFORM	Route <Route Name>: updated	Route updated due to changes in the Base System's geometry
INFORM	Route <Route Name>: Route with events deleted	Route in the Route Class deleted along with the remaining Event Points in an Event Class